

Tensor computations

# Finite mean angular condition number for identifiable tensor decomposition

**Difficulty:** challenging **Importance:** interesting to specialist**Status:** Solved

**Rating rationale:** Challenging because generic uniqueness does not supply the integrability estimates needed near degenerate decompositions; specialist importance reflects the particular angular condition number and volume-based input model.

## Resolution — 2026-09-11

**Affirmative resolution.** Matthew J. Colbrook's [complete manuscript, theorem in §1 and proof in §§2–4](#) proves finite mean angular condition number for every admissible format and every  $r \geq 3$  under the original generic complex identifiability assumption. The proof uses the exact volume-Gaussian input distribution and the derivative of the individually normalized summands. A bounded semialgebraic graph controls the angular derivative on the unit link; homogeneity then gives a finite radial integral. It does not assert a format-uniform bound or finite mean for the ordinary condition number.

The complete source passed [independent Codex-agent proof review](#). [Authorship, AI assistance and verification record](#). No external human peer review or formal verification is asserted. The original target below is retained verbatim; the difficulty, importance and rating rationale above are historical. This entry no longer contributes to the open count.

[Manuscript PDF](#). [Second independent review](#).

## Context and notation

Fix  $d \geq 3$ ,  $n_j \geq 2$ , and  $r \geq 2$ . Assume generic complex identifiability: outside a proper algebraic exceptional set, a complex tensor of rank  $r$  in this format has a unique unordered collection of  $r$  rank-one summands. Let  $M_r$  be the smooth identifiable locus of real rank- $r$  tensors, with induced Euclidean volume  $dV$ . The random input has density

$$d\mu(A) = Z^{-1} e^{-\|A\|_F^2/2} dV(A).$$

For the addition map  $\Phi(a_1, \dots, a_r) = \sum_i a_i$  on rank-one tensors, let  $\Psi$  be a local inverse at  $A$ . Use Frobenius norms and their product norm. Define

$$\kappa(A) = \|D\Psi(A)\|_2, \quad \kappa_{\text{ang}}(A) = \|D(p^{\times r} \circ \Psi)(A)\|_2, \quad p(a) = a/\|a\|_F.$$

Values on measure-zero exceptional sets do not affect the expectations. This samples tensors by volume, not independent Gaussian summands.

## Problem statement

Under this probability model, prove or disprove

$$\mathbb{E}_\mu \kappa_{\text{ang}}(A) < \infty$$

for every admissible format and  $r \geq 3$ . The rank-two case is a theorem. Normalizing each recovered summand inside the derivative is part of the definition; simply dividing the regular condition number by a scalar does not give this problem.

## Reference

Beltrán, Breiding, and Vannieuwenhoven, *The Average Condition Number of Most Tensor Rank Decomposition Problems Is Infinite*, *Foundations of Computational Mathematics* 23 (2023), 433–491: equation (6), Theorem 3, and Conjecture 2. The [author preprint](#) labels the conjecture 1.10.

## Status check — 2026-09-10

Rechecked the [journal version](#), [equation \(6\)](#), [Theorem 3](#) and [Conjecture 2](#), and searched for higher-rank angular-condition-number proofs. The theorem covers rank two, which the displayed target already excludes; no result proving finite mean for all admissible ranks at least three was located. The 2023 conjecture is the latest explicit statement checked, and a later unlocated result cannot be ruled out.